Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.18	20.15	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.02	-0.2% better
power (W), mean	626	644	+2.8% worse
power (W), peak	650	684	+5.3% worse
energy (Wh)	211	216	+2.7% worse
energy, parked workers still on at idle power (Wh)	211	216	+2.7% worse
CPU used (cores), mean	0.802	1.28	+60.1% better
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.033	0.053	+60.1% better
energy per core-hour (Wh)	781	502	-35.8% better
node-hours per core-hour	7.48	4.68	-37.5% better
pending pods, pod-minutes	0.533	0.267	-50.0% better
pending pods, peak	1.00	1.00	+0.0%
HPA replicas, mean	8.75	9.48	+8.3% worse
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.533	-31.9% better
requests timed	1,125	1,090	-3.1%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	67.94	105	+54.1% worse
response time (ms), median	45.30	89.85	+98.3% worse
response time (ms), 95th percentile	170	172	+1.0% worse
response time (ms), 99th percentile	252	218	-13.5% better
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	8.2548	5.1533	-37.6%	[-4.9713, -1.2315]	better
kwh_per_core_hour	0.8574	0.5490	-36.0%	[-0.4936, -0.1280]	better
utilisation	0.0331	0.0543	+64.0%	[+0.0079, +0.0359]	better
pending_pod_minutes_per_hour	1.5122	0.8399	-44.5%	[-4.5366, +1.8642]	no significant difference
hpa_shortfall_minutes_per_hour	2.2322	1.6798	-24.7%	[-3.5536, +2.5791]	no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.392 0.421 0.446 0.469 0.488 0.505 0.521 0.535 0.547 0.557 0.562 0.568 0.572 0.577 0.582 0.586 0.588 0.589 0.591
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	614	619	5.00	6.00	0.000	0.250	0.475	0.635

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	616	626	5.00	8.00	0.000	0.000	0.520	0.817
2	6.00	6.00	614	627	5.00	8.00	0.000	0.000	0.478	0.831
3	6.00	6.00	617	642	5.00	8.00	0.500	0.000	0.548	1.23
4	6.00	6.00	632	648	8.00	10.00	0.000	0.000	0.950	1.41
5	6.00	6.00	629	648	9.00	10.00	0.000	0.000	0.859	1.42
6	6.00	6.00	632	652	9.00	10.00	0.000	0.000	0.963	1.50
7	6.00	6.00	641	680	9.33	10.00	0.000	0.000	1.22	2.25
8	6.00	6.00	643	674	10.00	10.00	0.000	0.000	1.26	2.10
9	6.00	6.00	644	677	10.00	10.00	0.000	0.000	1.29	2.14
10	6.00	6.00	642	664	10.00	10.00	0.000	0.000	1.22	1.85
11	6.00	6.00	615	623	10.00	10.00	0.000	0.000	0.491	0.743
12	6.00	6.00	617	625	10.00	10.00	0.000	0.000	0.579	0.774
13	6.00	6.00	624	633	10.00	10.00	0.000	0.000	0.724	0.980
14	6.00	6.00	632	651	10.00	10.00	0.000	0.000	0.932	1.49
15	6.00	6.00	630	648	10.00	10.00	0.000	0.000	0.915	1.38
16	6.00	6.00	631	650	10.00	10.00	0.000	0.000	0.920	1.46
17	6.00	6.00	621	641	10.00	10.00	0.000	0.000	0.706	1.20
18	6.00	6.00	615	621	10.00	10.00	0.000	0.000	0.525	0.678
19	6.00	6.00	617	627	10.00	10.00	0.000	0.000	0.575	0.841

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.